

Zhuocheng (Tom) Qiao

AI Agent Engineer / AI Backend Engineer

Shanghai | M.S., Signal and Information Processing, East China Normal University (2022)

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[Engineering Portfolio](#): Agent Runtime, AI Harness, Open Source & Project Deep Dives



SUMMARY

Microsoft Software Engineer II with 4+ years building production AI agent runtimes, evaluation/agent harnesses, and backend platforms, with hands-on experience in distributed systems, productization, reliability, and cross-team delivery.

EXPERIENCE

Microsoft | Software Engineer II | Jun 2022–Present

Office Agent (Societas) | General-Purpose Agent Platform | Jul 2025–Feb 2026

A general-purpose agent for Microsoft 365 Personal and Family subscribers, reaching **27K WAU**—approximately **8.5%** of eligible users—within **two months** of global launch; led core work across Agent Runtime, productionization, reliability, and compliance engineering.

- **Native Tool Calling:** Drove implementation of the migration from text-based XML tool invocation to native function calling, with dual-mode rollout/rollback and full sub-agent-tree support for reliable cross-model execution.
- **Reliable Message Delivery:** Independently rebuilt long-running agent messaging with Redis Lists for 24-hour replayable state and Service Bus Sessions for retention, acknowledgement, redelivery, and per-session FIFO; introduced a rollout/rollback transport abstraction for backend interruptions and browser reconnect replay.
- **Agent Proxy:** Designed and delivered Agent Proxy to production from scratch, covering OBO authorization, task persistence and restart recovery, containerization, and CI/CD, enabling reliable external integration as a standalone service.
- **Data Security & Compliance:** Designed and implemented a user-level encryption scheme—hierarchical keys, hashed lookup, self-describing ciphertext, rotation, and gradual migration—later reused by partner modules; independently migrated Societas backend node-to-node traffic from HTTP to HTTPS through Key Vault certificate injection, PFX-to-PEM conversion, server certificate loading, and environment-wide cutover, and added CA verification to Agent Proxy downstream traffic, protecting data at rest and in transit.

Microsoft Copilot | Graph-based Agentic Loop & Production Engineering | Mar 2026–Present

Work spans Epichan's graph-based Harness and Loop Engineering practices, and Studio's production AI product and delivery systems.

- **Epichan (Contributing to Graph/Loop Engineering in Practice):** On Epichan's existing graph-based Harness (Developer / Reviewer / Judge AI agents scheduled along a DAG), designed and iterated the a11y-remediation-workflow Skill, codifying reproduction methods, evidence required across development/review/acceptance, and independent validation as reusable rules. Embedded the Loop Engineering principle of "fix the system, not just the task" as a domain Skill in the DAG, with mandatory checks for DOM/ARIA structures that pass while screen-reader behavior fails, misread keyboard-navigation semantics, and cross-scenario regressions from shared-event changes; incomplete evidence blocks advancement.
- **Studio Engineering & Production Reliability:** Work centered on Copilot's production inference pipeline—session ingress, Turn/Orchestrator scheduling, model-tool loops, safety/citation processing, and streaming responses; drove CI speed and stability through change classification, test parallelization, and automated flaky detection/isolation/tracking, reducing team build time by about **30%**; designed the hotfix-to-production path from scratch, optimized artifact promotion, and supported multi-region rollout and on-call.

Office Plus | Commercial Backend & Identity Governance | Jul 2024–Jun 2025

FY25 .NET/ABP commercial platform on Azure: **3.4M new paid subscriptions, \$6.9M revenue**; subscriptions, payments, entitlements, identity, and third-party delivery.

- **Plagiarism checking:** Led upload-to-refund workflows across orders, payments, entitlements, and Service Bus; the feature processed **6,320** orders in its first **10 days**.
- **Subscription and channel growth:** Led WeChat auto-renewal and Partner VIP top-up with decoupled contract/order/entitlement states, layered auth, idempotency, callback isolation, and compensation; after launch, auto-renewal reached **86%** of total orders; business data showed ARPPU **+58%** MoM, and manual renewal rose from **13.8%** to **24%**.
- **Identity and session governance:** Led soft deletion and refresh-token/JWT device-session modernization with recoverable state, audit, and scheduled cleanup; reduced record growth from **MAU × devices × 30** to **MAU × devices** with quotas, revocation, and gradual migration.

Earlier Projects: XPay (Jun–Oct 2022) | PC Manager Client (Nov 2022–Jun 2024)

TECHNICAL SKILLS

Backend/Languages: Python, C#/ .NET, Java, FastAPI, REST/SSE, Distributed Tasks & Asynchronous Processing

Agent Engineering: Function Calling, Multi-Agent/Sub-Agent, Agent Harnesses, Skills, Tool Orchestration, Context & Memory, Automated Evaluation & Quality Gates

Infrastructure: Azure, Service Bus, Redis, PostgreSQL, Docker, CI/CD, Logging & Observability

OPEN-SOURCE CONTRIBUTIONS

Contributed to open-source projects including Hermes Agent and Multica, with **10+ fixes** merged or incorporated upstream across agent security, runtime reliability, tenant isolation, and API robustness; experienced in issue discovery, root-cause analysis, patches, tests, and maintainer collaboration.